NEURAL WAVE | PROJECT DEFINITION

AI-based PV Detection from Electricity Consumption Profiles

Detection of photovoltaic installations, commissioning dates and nominal capacities using electricity consumption profiles only

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Challenge objective

Build a reproducible AI/ML solution that determines whether each customer has a photovoltaic installation and, when PV is detected, estimates its commissioning date and nominal installed capacity (kWp), using only the provided electricity consumption time series.

Dataset at a glance

60 labelled customer cases: 40 for development and 20 for hidden testing.
Additional data: 1148 unlabeled consumption time series.

1. Introduction to the Domain / Motivation for the Problem

AIL - Aziende Industriali di Lugano SA is a Swiss energy utility serving customers with very different electricity consumption profiles and supply points distributed throughout Switzerland. Effectively managing these customers requires understanding how their electricity withdrawals evolve over time.

The installation of a photovoltaic (PV) system at a customer site can structurally change the electricity consumption/withdrawal profile measured from the grid. During solar production hours, part of the customer demand can be covered locally by the PV system, reducing the amount of energy drawn from the grid. This change may leave recognizable patterns in the time series even when direct PV production measurements are not available.

For AIL, detecting these changes promptly means better understanding how customers' electricity withdrawals evolve and keeping information on energy installations at supply points up to date. Automatic detection of photovoltaic installations can support energy profile monitoring and operational processes, making use of existing data without requiring direct measurements of PV production.

The challenge explores how much can be inferred from the signal that is consistently available in this context: the customer electricity consumption time series. Weather, solar irradiation and geographical location data are not provided and must not be used.

AIL is also interested in approaches that can identify changes in consumption profiles without relying exclusively on labelled examples. This could support the future detection of other events, such as the installation of battery storage systems, for which no labelled cases are currently available.

Additional unlabelled consumption time series will be provided to support this exploration. Battery detection remains outside the scope and evaluation of this challenge.

Central question

How much can we infer about a photovoltaic installation - its presence, commissioning date and nominal capacity - by observing only the evolution of the customer electricity consumption profile?

2. Problem Statement

Participants must develop an automated pipeline that analyzes customer electricity consumption time series and produces three main outputs for each customer.

Task 1 - PV Presence Detection

Classify each customer as having or not having a photovoltaic installation.

- Required output: boolean TRUE/FALSE for each customer_id.

Task 2 - Commissioning Date Estimation

For customers classified as having PV, estimate the date from which the consumption profile becomes consistent with the initial PV system entering service.

- Required output: estimated date in YYYY-MM-DD format; empty if pv_installed = FALSE.

Task 3 - Nominal PV Capacity Estimation

For customers classified as having PV, estimate the nominal capacity of the initial installation in kWp using only the consumption profile and the provided time information. Irradiation and location data are not available.

- Required output: estimated capacity in kWp; empty if pv_installed = FALSE.

No specific algorithmic approach is required. The goal is to obtain a solution that generalizes to customers not included in the development dataset and can be executed reproducibly on the hidden test dataset.

Optional Task - PV Capacity Extension Detection

Some customers increase the capacity of their PV system after its initial installation, through one or more extensions. The labelled dataset includes three customers with extensions: one in the development set and two in the hidden test set. One of the test customers has two extension events after the initial installation.

Participants may optionally identify these events and estimate each extension date and the total nominal system capacity immediately after that extension. This task can earn up to 10 bonus points and is not required for the MVP.

For the three main tasks, the target date and capacity remain those of the initial installation. Subsequent extensions are evaluated separately.

3. Explanation of the Data

3.1 Dataset Overview

AIL provides 60 real, anonymized and labelled customer cases. Forty cases will be released to participants for development, training and validation, while 20 cases will remain hidden and will be used exclusively by the organizers for the final evaluation. The split takes customer diversity into account to retain variety in both datasets.

In addition, participants will receive 1148 unlabeled consumption time series with different historical coverage periods. Their use is optional.

Item	Description
Total labelled cases	60 anonymized customers
Released dataset	40 customers with time series and corresponding labels
Hidden test set	20 customers; time series will be used for final inference while labels remain private
Additional unlabelled data	1148 consumption time series with different historical coverage periods
Labelled data period	01.01.2022 - 31.12.2025
Time resolution	15 minutes
External information	No weather, location or solar irradiation data; the solution must rely on the provided data

3.2 Time Granularity and Daylight Saving Time

On a standard day, the dataset contains 96 intervals of 15 minutes. In Switzerland, daylight saving time transitions create two exceptions:

- 92 measurements on the day when clocks move forward by one hour (23-hour day);
- 100 measurements on the day when clocks move back by one hour (25-hour day).

The timestamp_epoch field is also provided to allow unambiguous chronological ordering, particularly around daylight saving time transitions.

3.3 Time-series File

The measurement file contains one row for each available time interval and uses the following fields:

Field	Type / unit	Description
customer_id	string / ID	An anonymized customer identifier. For labelled cases, it links time-series measurements to the corresponding customer labels.
energy_consumption_kwh	decimal / kWh	Electrical energy measured during the individual 15-minute interval. It represents the consumption/grid withdrawal recorded by the metering system during that interval.
timestamp_local	local datetime	Local date and time of the measurement in Switzerland (CET/CEST). It reflects daylight saving time transitions.
timestamp_epoch	Unix Epoch	Epoch timestamp representing the same instant, useful for unambiguous ordering of measurements, including during daylight saving time transitions.

Additional Unlabelled Time Series

The file energy_consumption_unlabelled.csv contains 1148 additional consumption time series. It uses the same fields, units and timestamp conventions as the measurement file above: customer_id, energy_consumption_kwh, timestamp_local and timestamp_epoch.

No corresponding labels file is provided. An absent label does not mean that PV is absent: these series have no validated information on PV presence, installation dates or capacities. Customers in the hidden test set will be excluded from this unlabelled dataset.

3.4 Labels File

For the 40 released cases, a separate file containing the ground-truth labels will be provided:

Field	Type / unit	Description
customer_id	string / ID	An anonymized customer identifier corresponding to the customer_id used in the time-series file.
pv_installed	boolean	TRUE if the customer has a photovoltaic installation during the observed period; FALSE otherwise.
pv_installation_date	date	Commissioning/installation date of the initial PV system. The field is empty when pv_installed = FALSE.
pv_capacity_kwp	decimal / kWp	Nominal capacity of the initial PV installation, before any extensions. The field is empty when pv_installed = FALSE.

PV Extension Labels

The file pv_extensions_labels_dev.csv contains only confirmed PV extension events, with one row per event. A customer may appear multiple times if more than one extension occurred.

Field	Type / unit	Description
customer_id	string / ID	An anonymized customer identifier matching the customer_id in the time-series and initial-installation labels files.
extension_date	date	Commissioning date of this PV capacity extension, in YYYY-MM-DD format.
pv_capacity_after_extension_kwp	decimal / kWp	Total nominal PV capacity immediately after this extension, including the initial installation and any previous extensions.

Customers not listed in pv_extensions_labels_dev.csv have no labelled extension events. For the three main tasks, pv_installation_date and pv_capacity_kwp refer to the initial installation. Extensions are evaluated separately using the total nominal capacity after each event, not the added capacity. All hidden test labels, including extension labels, remain private.

3.5 Data Usage Constraints

- The solution must rely on the provided consumption time series and associated time information. Development labels and additional unlabelled time series may be used; use of the unlabelled data is optional.
- Weather, geographical location and solar irradiation data are not provided and must not be used.
- Calendar features may be derived from the provided timestamps, for example hour of day, day of week, month and seasonal patterns.
- The data are anonymized; customer_id does not contain identifying information about the customer.
- Attempts to reidentify customers or supply points are prohibited.

4. Project Evaluation

Final evaluation will be performed by the organizers on the 20 hidden test cases. The test dataset will use the same time-series schema as the released dataset, but the ground-truth labels will not be shared with participants.

Each team must submit an inference.py entry point together with any required model artifacts and minimal execution instructions. The script must read the test CSV from the specified path, perform all required preprocessing automatically and produce the two CSV files specified below.

4.1 Evaluation Output Format

Write both output files to the current working directory, using UTF-8 encoding, semicolons separators, a decimal point and the exact column names and order shown below. Dates must use YYYY-MM-DD, booleans must be TRUE/FALSE, and customer_id values must be preserved as strings, including any leading zeros. Use empty fields for values that do not apply.

Main Output - predictions.csv

Include exactly one row for every customer_id in the test CSV. Date and capacity describe the initial installation and must be empty when pv_installed = FALSE.

Output field	Required content
customer_id	Customer identifier present in the test set.
pv_installed	Boolean prediction TRUE/FALSE.
pv_installation_date	Estimated initial commissioning date in YYYY-MM-DD format; empty if pv_installed = FALSE.
pv_capacity_kwp	Estimated nominal capacity of the initial installation in kWp; empty if pv_installed = FALSE.

Extension Output - pv_extensions_predictions.csv

Include one row per predicted extension. A customer may appear on multiple rows. The capacity is the total nominal system capacity immediately after the corresponding event. If no extensions are predicted or the optional task is not implemented, create this file with its header only.

Output field	Required content
customer_id	Customer identifier present in the test set.
extension_date	Estimated extension date in YYYY-MM-DD format.
pv_capacity_after_extension_kwp	Total nominal system capacity immediately after this extension, in kWp; not the added capacity.

Synthetic Output Examples

The following examples are fictitious. DEMO_001 starts at 10 kWp and reaches 15 kWp, then 18 kWp, after two extensions. DEMO_002 has no PV.

predictions.csv

```
customer_id,pv_installed,pv_installation_date,pv_capacity_kwp
DEMO_001,TRUE,2022-05-01,10
DEMO_002,FALSE,,
```

pv_extensions_predictions.csv

```
customer_id,extension_date,pv_capacity_after_extension_kwp
DEMO_001,2023-06-01,15
DEMO_001,2024-07-01,18
```

4.2 Main Score

Main scoring formula

Main score = F1 × (70 + 15 × D + 15 × P)

The main score ranges from 0 to 100. All indicators F1, D and P range from 0 to 1.

F1 - PV Detection

F1 combines precision and recall for the pv_installed classification. It penalizes both false positives and false negatives. F1 = 1 represents perfect classification.

D - Commissioning Date Accuracy

For each correctly detected PV installation, the estimated date is compared with the true date. The score for each case decreases linearly with the absolute error in days:

- 0 days error = 1.00
- 15 days error = 0.50
- 30 days or more error = 0.00

Formally: D_i = max(0, 1 - |error_days| / 30). D is the mean of D_i over the relevant correctly detected PV cases. A missing or invalid date receives 0 for the D component without removing the case from the mean.

P - Nominal Capacity Accuracy

For each correctly detected PV installation, the estimated capacity is compared with the true nominal capacity. The score for each case decreases linearly with the absolute percentage error:

- 0% error = 1.00
- 25% error = 0.50
- 50% or more error = 0.00

Formally: P_i = max(0, 1 - absolute_percentage_error / 50%). P is the mean of P_i over the relevant correctly detected PV cases. A missing, non-numeric or invalid capacity receives 0 for the P component without removing the case from the mean.

Main Score Interpretation

PV presence detection contributes up to 70 points. Date accuracy and capacity accuracy contribute up to 15 points each. The entire result is also multiplied by F1, so strong date and capacity estimates for only a small number of installations are not enough to obtain a high score when overall detection performance is weak.

Example

With F1 = 0.90, D = 0.80 and P = 0.70:
Main score = 0.90 × (70 + 12 + 10.5) = 83.25 points.

If no PV installation is correctly detected, the main score is 0.

4.3 Optional PV Extension Bonus

Up to 10 additional points

Bonus = F1_ext × (7 + 1.5 × D_ext + 1.5 × P_ext)
Final score = main score + bonus, for a maximum of 110 points.

F1_ext evaluates individual events across the entire hidden test set: F1_ext = 2TP / (2TP + FP + FN), where TP are matched extensions, FP are false alarms and FN are missed extensions. Match events for the same customer within ±30 days, one to one. Maximize the number of valid matches, then minimize total date error. Unmatched or duplicate predictions count as false alarms.

D_ext and P_ext are means over correctly matched events, using the main-score rules: the date score decreases linearly from 1 for an exact date to 0 at 30 days of error; the capacity score decreases linearly from 1 for an exact estimate to 0 at 50% absolute percentage error. P_ext compares the predicted and true total nominal system capacity immediately after each extension, not the added capacity. It is independent of the predicted initial capacity.

A missing or invalid date prevents matching. A missing, non-numeric or non-positive capacity scores 0 in P_ext without removing the matched event from the mean.

If no extension is matched, or the optional task is not implemented, the bonus is 0. Extension prediction errors do not deduct points from the main score. Ties in the final score are resolved by the higher main score.

4.4 Inference Script Requirements

Run the following command from the submission folder:

```
python inference.py path/to/test_set.csv
```

1. Provide inference.py as the single entry point. It must accept exactly one positional argument: the path to the test time-series CSV file, not a folder.
2. Include all preprocessing required for inference.
3. Load any submitted model weights or artifacts automatically. Supporting Python modules and local artifacts may be included alongside inference.py.
4. Write predictions.csv and pv_extensions_predictions.csv to the current working directory in the formats specified in Section 4.1. The main file must contain exactly one row per customer; the extension file may contain only its header.
5. Once dependencies have been installed, complete inference without manual intervention or external network connections.
6. Include a concise README describing dependencies, environment setup and execution instructions.

5. Potential Solution / MVP

Participants may use the 40 labelled cases and the 1148 unlabeled time series to develop supervised, unsupervised, semi-supervised or hybrid approaches.

Use of the unlabelled time series and detection of PV capacity extensions are optional. The MVP requirements remain unchanged. Battery detection is outside the scope of the challenge.

Expected MVP

1. Automatically import and validate the provided data.

2. Correctly handle the 15-minute temporal structure, including days with 92 or 100 measurements due to daylight saving time transitions.

3. Produce a pv_installed classification for each customer.

4. For customers classified as having PV, estimate the initial commissioning date and initial nominal capacity in kWp.

5. Generate the standardized output required for evaluation.

6. Run through a script without manual steps on the hidden test set.

A graphical user interface is not required. Visualizations or dashboards may be created to support the final presentation, but they do not directly contribute to the quantitative score defined above.

6. Additional Information

6.1 Domain Concepts

Concept	Explanation
Consumption profile / load profile	A time series describing how much electrical energy is recorded for a customer over time.
kWh	A unit of energy. In this dataset it represents the energy measured in each 15-minute interval.
Photovoltaic system (PV)	A system that produces electricity from solar radiation. When locally generated energy is self-consumed, the customer grid withdrawal may decrease.
kWp	Kilowatt-peak: the nominal peak capacity of a photovoltaic installation under standard test conditions.
Commissioning date	The date from which the PV installation starts to influence the observed electricity consumption behavior.
Change point	A point in time when the statistical or behavioral characteristics of a time series change persistently.

6.2 Assumptions and Limitations

- Direct photovoltaic production measurements are not available.
- Weather, solar irradiation and geographical location data are not provided and must not be used.
- Nominal PV capacity is not directly observable from the electricity consumption profile alone; capacity estimation is therefore an inference task and will be evaluated as such.
- Labelled time series may have different coverage within 01.01.2022-31.12.2025. The additional unlabelled time series have their own historical coverage periods.

6.3 Tools and Dependencies

No access to proprietary AIL frameworks or APIs is required. Participants may use open-source libraries and frameworks of their choice. The final solution must be reproducible in the evaluation environment defined by the organizers and must not depend on external weather, location or solar irradiation data.

LLMs and Coding Agents

- **Development assistance: LLMs and coding agents may be used to write, explain, debug and document code.**
- **Submitted solution: No LLM may be part of the analysis or prediction pipeline, whether accessed remotely or run locally.**
- **AIL data: AIL datasets, labels and extracts must not be uploaded, transmitted or made accessible to external LLM services or coding agents, even if anonymized.**